

Modeling and analysis of solvent removal during Li-ion battery electrode drying

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In this work, we study the design aspects and process dynamics of solvent removal from Lithium-ion battery electrode coatings. For this, we use a continuum level mathematical model to describe the physical phenomenon of cathode drying involving coupled simultaneous heat and mass transfer with phase change. Our results indicate that around 90% of solvent is removed in less than half of the overall drying time. We study the effect of varying temperature and air velocity on the drying process. We show that the overall drying energy can be reduced by at least 50% by using a multi-zone drying process. Also, the peak solvent flux can be reduced by at least 40%. We further present the effect of using an aqueous solvent instead of N-Methyl-2-pyrrolidone (NMP) in electrode drying. Our results indicate that Water dries nearly 4.5 times faster as compared to NMP and requires nearly 10 times less overall drying energy per kg of solvent.

Keywords: drying, mathematical modeling, Li-ion battery cathode, heat and mass transfer

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Introduction

Lithium-ion batteries are becoming increasingly popular for their use in a variety of consumer applications and especially, in hybrid and electric vehicles. Consequently, the development of low cost and high energy density lithium-ion batteries is receiving ever greater interest from both academia and industry. The research effort in developing efficient and affordable batteries can be broadly classified into three main areas of interest. They are increasing energy density, ensuring operational safety, and reducing cost. These objectives are mainly addressed in the studies developing new active materials and benchmarking process control practices. In this regard, optimization of battery manufacturing operations and the in-depth understanding of underlying physical/chemical processes have a great potential to offer crucial insights in addressing some of the prevalent problems in battery research, especially in improving quality and cost.

One of the major components of the overall lithium-ion battery cost is the processing cost of an electrode[1]. The physical properties of an electrode coating, such as coating structure and the distribution of material components within the coating layer (including active materials, additives, and polymer binders), greatly influence its electrochemical performance [2-6]. The production of an electrode involves a series of steps including mixing of materials, coating of material-solvent slurry on a current collector, drying of electrode coatings, and then calendaring of dried coatings. A number of publications [4, 5] study the impact of slurry mixing on the quality of electrodes. Also, some studies address the effect of calendaring [3, 7] and coating methods [8, 9] on the electrochemical performance of electrodes. However, studies focusing on the analysis of drying process and its impact on the cost [10] and quality of an electrode are limited.

The process of drying has a significant impact on: (i) the distribution of materials within the coating due to the interplay of various forces (such as capillary action, diffusion, and convection), and (ii) the structure of the porous coatings. In addition, drying as a unit operation is inherently slow, expensive, and energy intensive [10]. Thus, the study of drying process in electrode coatings is of great importance (i) to understand the interplay of mass transport and heat transfer processes, particularly if they can be experimentally validated and correlated to electrode quality, and (ii) to extract insights on improving the processing speed and energy demand. Consequently, the study of electrode drying and its effect on electrode quality (and cost) is receiving an increasing attention from the research community [1, 10, 11].

Drying of coatings in general has been studied for several decades in a variety of industries including paper, paint, textile, food, agriculture, wood, ceramics, pharmaceutical products, and wastewater sludge. A variety of mathematical models have been proposed to explain the process and phases of drying [2, 12-14]. Over the last few decades, the concept of simultaneous heat and mass transfer processes has been widely adopted to explain drying in various coatings or materials. Defraeye [13] presents a comprehensive account of various approaches and models proposed so far to study drying in a variety of systems. The drying of porous coatings and its impact on the coating properties has been extensively studied, particularly, in the paper industry [12, 13, 15, 16]. In contrast, similar research efforts in electrode coatings with application to lithium-ion batteries are limited. Although, both of these applications have similarity, there are significant differences that require them to be studied separately. However, findings from one can very well be adopted as a basis to develop models for the other[17]. Gutoff [18, 19] proposed a series of simple models

considering simultaneous heat and mass transfer for drying of coatings on continuous coated webs. He studied the rate of drying and temperature profile in the coating through an iterative algorithm implemented on an excel worksheet. Kim and Rhee[20] presented the effect of using near infrared (NIR) drying method instead of the conventional oven drying on the electrochemical properties of cathodes. The study of binder particle re-distribution within the coating layer during drying of electrode has attracted a significant attention in the literature [17, 21-24]. This is because the distribution profile of binder particles within coating layer has a significant impact on the quality of coating[25]. Li et al.[25] studied the distribution of binder, Polyvinylidene fluoride (PVDF), in LiCoO₂ based coatings in aqueous and organic [N methyl-2-pyrrolidone (NMP)] mediums during drying to highlight its impact on the electrochemical performance of the resulting electrode. Lim et al.[26] studied the development of stress during the drying of anodes coated with the graphite and sodium carboxymethyl cellulose (CMC) or styrene butadiene rubber (SBR) slurry. Westphal et al.[27] showed the effect of process parameters such as temperature and nozzle speed on the elasticity, electrical volume resistivity and adhesion strength between electrode coating and current collector. Similarly, Baunach et al.[2] studied the impact of drying rate on the coating adhesion of a graphite based anode. They highlighted that the quality of electrode can be controlled by solving an optimization problem with a tradeoff between drying rate and cost. Jaiser et al.[14] studied the phenomenon of film solidification in lithium-ion battery anodes. They investigated the impact of slurry conditions and process parameters during drying on the adhesion of coated film on the current collector. Jaiser et al.[28] proposed a strategy to reduce time for drying time and maintain the quality of the coating through a 3-stage drying process. They vary the rate of drying in the three stages by varying temperature

and convective flow of moisture carrier. They defined characteristic points for changing the drying rates based on the amount of moisture present in the coating. Li et al.[29] investigated the impact of residual moisture after drying on the electrochemical performance of electrodes. They compared the performance of a $LiNi_{0.5}Mn_{0.3}Co_{0.2}O_2$ (NMC532) cathode in both aqueous and NMP solvent after drying. They showed that the residual moisture in both the cases are comparable and in general lower residual moisture favored coulombic efficiency in the first cycle. Schunemann et al.[30] identified that apart from the material costs, drying contributes the maximum cost to the overall cost of a Li-ion based cell and showed the potential savings in cost by increasing the solids content (or reducing solvent) in the slurry making process. They further highlighted drying is also a very high energy consuming step in the battery manufacturing process. Ahmed et al.[10] developed a process model to estimate the cost and energy demand of drying the cathode and recovering the solvent (NMP). They showed that the energy demand for drying is over 10 kWh per kg of NMP vaporized for the given plant capacity, which is about 45 times in excess to the actual heat required for vaporization of equivalent solvent. Also, the authors estimated that the cost of drying and recovery of solvent amount to about 3.4% of the overall battery cost.

The aforementioned review of existing literature highlights an increasing interest in studying the drying of electrodes for lithium-ion battery application, especially, in the last couple of years. However, most of these publications [14, 17, 23, 24, 28] focus on studying the change in binder concentration profile within the coating layer during drying and then associating this phenomenon to the quality of resulting electrode. Other publications [1, 10, 11] have focused on the energy demand and economic aspects of drying. Clearly, few works focus on studying the process of electrode drying by understanding the mass transport and

heat transfer processes within the coating layer. Also, the effect of different solvents (aqueous and organic) on drying process has not been studied. This forms the primary focus of our work.

In this paper, we develop a mathematical model to study and understand the physical processes such as mass transport and heat transfer during drying of Li-ion battery electrode coating. We present a set of cases to study solvent drying in electrode coatings during the manufacturing process of battery electrodes. We further present an analysis on the effect of various (process and system) parameters on the drying of electrodes. We then discuss potential causes that can lead to quality defects and present an analysis of strategies to mitigate such issues. Finally, we study the effect of using aqueous versus organic solvents on drying.

Drying process model

The drying of porous electrode coatings is inherently very complex and involves simultaneous heat and mass transfers in all three phases (solid, liquid, and gas) represented by transport equations based on conservation laws for mass and energy. We follow the volume averaging technique proposed by Whitaker[31, 32] and popularized by his co-workers[33] and others[12, 18, 33-35] to develop a macro-scale model for drying. Here, the porous material is assumed to be in a continuum and the dependent variable (such as liquid phase density or gas phase density) is averaged over a small representative volume consisting of all three phases[34, 35]. Now, every point in the coating layer is assumed to be the center of such a representative volume, which can change composition (for liquid and gas phases) based on space and time. The volume averaging technique and the derivation of

the transport equations for drying have been presented by a number of articles in the literature[31, 32].

An essential factor in modeling is to understand the process conditions where the formulated equations and assumptions are physically valid. In other words, it is essential to define the scope of a mathematical model with respect to process/physical conditions of the system (the drying material). In general, the drying literature has classified drying operation into a number of sub-categories or regimes based on solvent concentration, drying rate, and moisture content. The process of drying porous coatings and solids, typically, consists of two phases. In the first phase, the solid particles are suspended in a pool of solvent and drying is mainly a surface phenomenon. Here, the rate of solvent drying is dependent only on the rate of evaporation at the surface of solvent-solid reservoir. As long as the process conditions remain constant, the rate of drying in this phase remains constant and so, is also referred to as the constant rate period. This first drying phase concludes as the solid particles come closer to form a porous structure. This porous structure is completely saturated with the liquid phase solvent. As drying proceeds the porous structure is no longer saturated with liquid phase solvent alone as the pore volume is now filled with both vapor phase and liquid phase, as shown in Fig. 1. At this time, the rate of drying is governed by the complex interplay of multiple physical phenomena within the porous electrode structure involving solvent evaporation, mass transport, and simultaneous heat and mass transfer. Clearly, the second phase of drying is significantly longer and highly complex as compared to the first phase. Thus, this regime of drying forms the focus of our study. Another important feature (especially, electrode coatings) is the rigidity of the porous structure, i.e. whether the coating deforms or shrinks during the drying process. The shrinkage of coating mainly happens

during the first phase of drying, where the solid-solvent slurry shrinks to form a porous structure. In the second phase of drying, the porous coating further shrinks as the liquid phase solvent evaporates and is replaced by the drying medium (in this case ambient air). However, the shrinkage of coating thickness in the second phase of drying is relatively less as compared to the first phase.

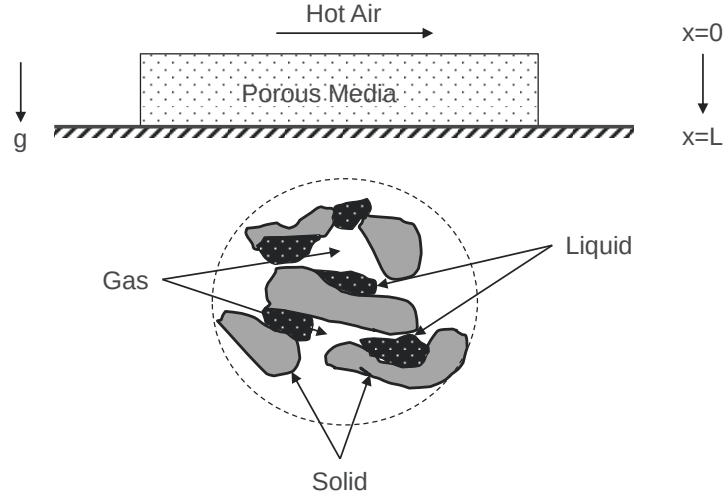


Figure 1 Physical model of drying and volume averaging element

In this work, we employ a volume averaging method to model the aforementioned drying process and represent the solvent loaded electrode coating as an unsaturated porous media. In other words, we model the electrode drying in the second phase of drying as described earlier, where the pore volume in the porous electrode structure is filled with both liquid and vapor phase as shown in Fig. 1. The electrode coating is dried by a drying medium such as hot dry air flowing over the top of the electrode coating. The temperature of the drying medium, in this case hot dry air, is not higher than the saturation temperature (i.e. boiling point) of the solvent at atmospheric pressure. Since the main focus of this study is to establish a basic understanding of solvent drying phenomenon within electrode coating, our interest was limited to capture the transient behavior in x -direction (i.e. thickness of

coating). Thus, we restrict ourselves to the one-dimensional model of the drying process. The following additional assumptions were made to keep the model simple and computationally tractable.

- 1) Heat and mass transfer are one dimensional
- 2) The solvent does not have any binder/polymer
- 3) The solid phase is homogeneous throughout the porous media
- 4) Both the liquid and air/vapor phases are continuous
- 5) Darcy's law holds for the gas and liquid phases
- 6) The binary air/vapor mixture behaves like an ideal gas and Fick's law can be applied to describe air-vapor diffusion
- 7) Energy transfer is due to conduction in all three phases and convection in liquid and gas phases. Thermal properties, such as specific heat and thermal conductivity, remains constant within the temperature range of interest

Mathematical formulation

The electrode drying system consists of solid, liquid, and gas phases. Let ε_s denote the volume fraction of solids in the system. So, $\varepsilon = 1 - \varepsilon_s$ represents the volume fraction of the void space (or porosity). This void space, is filled with both liquid and gas. Here, we define saturation (S) as the volume fraction of liquid within the void space [12, 15, 16]. Thus, saturation is given as

$$S = \frac{\text{volume of liquid}}{\text{volume of void space}}$$

Note that $(1 - S)$ represents the volume fraction of gas phase in the void space. Then, the overall volume fraction for liquid and gas phases are εS and $\varepsilon(1 - S)$, respectively. Recall

that, in this study, we model the electrode coating as an unsaturated porous media, i.e. volume of liquid phase solvent is less than the pore/void space. Thus, our study is limited to the case with $S < 1$.

Now, the mathematical model mainly consists of mass and energy conservation equations. The solvent mass conservation equation is written for both liquid and vapor phase separately. The liquid phase mass conservation equation is,

$$\frac{\partial}{\partial t}(\rho_l \varepsilon S) + \frac{\partial}{\partial x}(\rho_l \varepsilon S u_l) = -\dot{m} \quad (1)$$

where, ρ_l denotes the density of the liquid phase, $\dot{m}$ is the mass rate of evaporation, and u_l is the average flux velocity of liquid phase. The negative sign with $\dot{m}$ denotes the ‘sink’ or loss term for liquid-phase solvent in the balance equation. Note that the density of liquid is assumed constant within the temperature range of interest. The liquid phase velocity is calculated from Darcy’s equation for fluid flow through porous medium, and is given as

$$u_l = -\frac{k_l K}{\mu_l \varepsilon S} \left(\frac{\partial p_l}{\partial x} - \rho_l g \right) \quad (2)$$

where, $k_l K$ denote the permeability of liquid in the porous coating, μ_l is the liquid phase viscosity of the solvent, and p_l is the pressure of liquid phase.

In the gas phase, the solvent vapor mass transport equation is given as follows.

$$\frac{\partial}{\partial t}(\rho_v \varepsilon (1 - S)) + \frac{\partial}{\partial x}(\rho_v \varepsilon (1 - S) u_v) = \dot{m} \quad (3)$$

where, ρ_v is the density of the vapor phase and u_v is vapor phase velocity. Note that $\dot{m}$ has a positive sign and denotes the generation term in the balance equation. The flux velocity is due to (i) convective flow of gas through pores, and (ii) diffusion of vapor into air. The gas phase flow (u_g) is calculated from the Darcy’s flux equation while, the diffusive flow is

computed from Fick's law. The overall (convective + diffusive) flow of vapor phase is then given as the following[12].

$$u_v = u_g - \frac{\rho_g M_a M_v D}{\rho_v M_g^2} \frac{\partial}{\partial x} \left(\frac{p_v}{p_g} \right) \quad (4)$$

$$u_g = - \frac{k_g K}{\mu_g \varepsilon (1 - S)} \left(\frac{\partial p_g}{\partial x} - \rho_g g \right) \quad (5)$$

where, ρ_g is the density of the gas phase, M_a , M_v , M_g are molecular weights for air, solvent, and solvent-air mixture, D is the diffusivity of solvent vapor in air, p_v is the partial pressure of the vapor, p_g denotes the total pressure of gas phase, $k_g K$ denotes gas phase permeability, and μ_g is gas phase viscosity.

Apart from the solvent vapor, the gas phase also consists of dry unsaturated air. So, we write the following conservation equation for air in the gas phase.

$$\frac{\partial}{\partial t} (\rho_a \varepsilon (1 - S)) + \frac{\partial}{\partial x} (\rho_a \varepsilon (1 - S) u_a) = 0 \quad (6)$$

where, ρ_a is density of air and u_a is air flux calculated (similar to vapor velocity) from Darcy's equation for convective flow (u_g) and Fick's law for diffusive flow. The air velocity is given as the following.

$$u_a = u_g - \frac{\rho_g M_a M_v D}{\rho_a M_g^2} \frac{\partial}{\partial x} \left(\frac{p_a}{p_g} \right) \quad (7)$$

where, p_a is partial pressure of air and u_g is gas flow determined from darcy's law (Eq. 5).

The equation for the conservation of energy is given as follows.

$$\begin{aligned} & (\rho_m C_{Pm}) \frac{\partial T}{\partial t} + h_{vap} \dot{m} + \\ & (\rho_l \varepsilon S u_l C_{Pl} + \rho_v \varepsilon (1 - S) u_v C_{Pv} + \rho_a \varepsilon (1 - S) u_a C_{Pa}) \frac{\partial T}{\partial x} = \frac{\partial}{\partial x} \left(\lambda \frac{\partial T}{\partial x} \right) \end{aligned} \quad (8)$$

$$\rho_m C_{pm} = \rho_s C_{ps}(1 - \varepsilon) + \rho_l C_{pl}\varepsilon S + \rho_v C_{pv}\varepsilon(1 - S) + \rho_a C_{pa}\varepsilon(1 - S) \quad (9)$$

where, T is temperature, h_{vap} is enthalpy of vaporization, C_p is specific heat capacity, ρ is density, and $\lambda (= (1 - \varepsilon)\lambda_s + \varepsilon S\lambda_l + \varepsilon(1 - S)\lambda_g)$ is the thermal conductivity. The subscripts s, l, v, a, g represent different phases of solid, liquid, vapor, air, and gas, respectively.

The liquid phase mass transport is also assisted by capillary force within the pore space. This capillary pressure (p_c) is computed using the Leverett's formula as following.

$$p_c = p_g - p_l = \sqrt{\frac{\varepsilon}{K}} f(S) \sigma \quad (10)$$

where, K is the saturated permeability of the porous coating and σ is the temperature dependent surface tension of solvent. $f(S)$ and σ are given as follows.

$$\sigma(T) = \sigma_0 - \beta T \quad (11)$$

$$f(S) = 0.364(1 - e^{-40(1-S)}) + 0.221(1 - S) + 0.005/(S - S_f) \quad (12)$$

where, β is thermal coefficient for surface tension and S_f is the final or irreducible moisture content of the porous material.

The saturated permeability (K) of the bed can be estimated using the Carman-Kozeny equation. Also, the gas and liquid phase permeabilities are dependent on S . So, we write

$$K = \frac{d_p^2 \varepsilon^3}{180(1 - \varepsilon)^2} \quad (13)$$

$$k_l = \left(\frac{S - S_f}{1 - S_f} \right)^3 \quad (14)$$

$$k_g = 1 - 1.1S \quad (15)$$

where, d_p is the average particle diameter.

From ideal gas law, we have the following.

$$\rho_v = \frac{p_v M_v}{RT} \quad (16)$$

$$\rho_a = \frac{p_a M_a}{RT} \quad (17)$$

Let V_0 be the initial volume of the electrode coating and θ be the weight fraction of electrode particles in the electrode-solvent slurry. Given the initial thickness of the wet coating (h_0), the final thickness of the dried coating (h_f), and initial solvent saturation (S_0), we can approximate the initial and final porosities (ε_0 and ε_f), and the final solvent saturation (S_f) assuming 99.9% by volume of solvent is removed, as follows.

$$\text{Weight of the coating (solids + liquid), } w_{film} = \frac{V_0}{\left(\frac{\theta}{\rho_s} + \frac{1-\theta}{S_0 \rho_l}\right)}$$

where, $\left(\frac{1-\theta}{S_0 \rho_l}\right)$ represents void volume, V_{void} , and $\frac{\theta}{\rho_s}$ represents solids volume, V_{solids} .

$$\text{Initial porosity, } \varepsilon_0 = \frac{V_{void}}{V_0}$$

$$\text{Final porosity, } \varepsilon_f = \frac{V_{void}}{V_f}$$

$$\text{Final saturation, } S_f = \frac{0.001 * S_0 * V_{void}}{\varepsilon_f * V_f}$$

where, $V_f (= \frac{h_f}{h_0} V_0)$ is the final volume of the electrode film.

As mentioned earlier, electrode coatings shrink considerably during the drying process. To account for this change in volume of coating during drying, we assume that porosity (ε) varies linearly with saturation (S).

$$\varepsilon = \varepsilon_0 - \left(\frac{\varepsilon_0 - \varepsilon_f}{S_0 - S_f}\right)(S_0 - S) \quad (18)$$

To this end, we have 5 unknowns (i.e. $S, p_v, p_a, T, \dot{m}$) and 4 equations (i.e. Eqs. 1, 3, 6, and 8). Here, $\dot{m}$ is dependent on both the saturation (S) and temperature (T). Thus, we compute the mass rate of evaporation, $\dot{m}$ using the following expression [36].

$$\dot{m} = \frac{k_m M_v}{RT} a (S - S_f) (p_{v_{eq}} - p_v) \quad (19)$$

where, k_m is the mass transfer coefficient, $p_{v_{eq}}$ is the equilibrium vapor pressure of the solvent, and a is the evaporating area per unit volume. In principle, a is a non-linear function of pore volume, porosity, sphericity of particles, and solvent saturation S . We approximate a as a function of porosity, volume, and saturation. For this, we assume that the change in area is a function of porosity and volume change is only associated with the change in coating thickness ($h_0 - h_f$) during drying. We use the following relation to approximate a .

$$a = \frac{1}{h_0 - \frac{h_0 - h_f}{\varepsilon_0 - \varepsilon_f} (\varepsilon_0 - \varepsilon)} \quad (20)$$

Note that Eq. 19 follows our assumption that the rate of drying is fast enough that the vapor pressure may approach the equilibrium vapor pressure.

Next, we specify the boundary conditions to our drying problem. Since we consider a one dimensional model, we only need to specify two system boundaries. As shown in Fig.2, they are (i) at the impermeable coating-current collector interphase, i.e. $x = L$, and (ii) at the exposed air-coating interphase, i.e. $x = 0$.

Intuitively, we impose no flux condition for mass transport across the impermeable boundary. Now, the current collector foil has heat capacity of its own and is also a good conductor of heat. So, in principle, there is a possibility of heat loss across the impermeable boundary due to the heat capacity of the foil and convection on the other side. As the drying

chamber is typically enclosed in industrial operations so, the convective losses can be ignored with no loss of generality. Furthermore, most modern equipment coat and dry on both sides of the electrode simultaneously. In such a scenarios, the current collector foil is simultaneously heated from both sides. So, the temperature loss due to the heat capacity of this current collector foil is greatly reduced. Also, as we show later that the heat transfer across the electrode coating is very fast so, we ignore the heat loss associated with current collector heat capacity at $x = L$ for the sake of simplicity. Thus, at $x = L$, we impose that heat flux is zero. To reflect this at $x = L$, we write the following equations for the flux of solvent, air, and heat, respectively.

$$\rho_l \varepsilon S u_l = 0 \quad (21)$$

$$\rho_v \varepsilon (1 - S) u_v = 0 \quad (22)$$

$$\rho_a \varepsilon (1 - S) u_a = 0 \quad (23)$$

$$\frac{\partial T}{\partial x} = 0 \quad (24)$$

At the exposed interface of air-coating ($x = 0$), we enforce natural (or continuous) boundary conditions for mass and heat fluxes. Also, the pressure at the interface is constant and equal to the atmospheric pressure.

$$\rho_l \varepsilon S u_l + \rho_v \varepsilon (1 - S) u_v = \frac{k_m M_v}{RT} (p_v - p_{v_{eq}}) \quad (25)$$

$$p_g = p_{atm} \quad (26)$$

$$\frac{\partial T}{\partial x} = k_T (T - T^0) \quad (27)$$

where, $p_{v_{eq}}$ is the equilibrium vapor pressure of solvent, k_m is the mass transfer coefficient at the exposed interface ($x = 0$) and k_T denotes the heat transfer coefficient.

Finally, we define the following initial conditions for the three independent variables temperature, pressure, and moisture distribution.

$$S = S_0 \quad (28)$$

$$p_g(x, 0) = p_{atm} \quad (29)$$

$$T(x, 0) = T_{initial} \quad (30)$$

This completes our drying model with three independent variables liquid saturation (S), partial pressures for vapor and air (p_v, p_a), and temperature (T) comprising governing equations (Eqs. 6, 7, and 9), supporting equations (Eqs. 2, 4, 5, 8, 9, 11–20), boundary conditions (Eqs. 21–26), and initial conditions (Eqs. 27–29).

Results and discussion

The aforementioned model was solved using FlexPDE 7.0, which uses finite element method to solve the coupled partial differential equations. All physical parameters used in our simulation are listed in Table 1. The results obtained from solving the model is discussed and analyzed as follows.

Table 1. Model parameters for base case scenario

Parameters	
Cathode material	NMC532
Thickness	158 microns
Particle size	10 microns
Heat transfer coefficient	100 $\text{Wm}^{-2}\text{K}^{-1}$
Mass transfer coefficient	0.01 ms^{-1}
Solvent @ T = 368.15 K	NMP
Surface tension	6.13E-05 Nm^{-1}
Diffusivity	1.00E-05 m^2s^{-1}
Molecular weight	99 kgkmol^{-1}
Viscosity	0.000686 Pas
Density	824 kgm^{-3}
Enthalpy of vaporization	513 kJkg^{-1}
Thermal conductivity	1.21E-04 $\text{kWm}^{-1}\text{K}^{-1}$
Specific heat capacity	1.9 $\text{kJkg}^{-1}\text{K}^{-1}$

Base scenario

We study the drying of a Lithium Nickel Manganese Cobalt (NMC-532) oxide based cathode in N-Methyl-2-pyrrolidone (NMP) solvent in a convective dryer supplied with hot dry air at a temperature of 368.15 K (95 °C). For this, we assume that the slurry of cathode material and NMP solvent consists of $\approx 60\%$ by weight of solid particles. Also, the initial thickness of the coating (also known as wet gap) is about 158 μm . As mentioned earlier, we assume the electrode coating to be dried if 99.9% by volume of solvent is removed. We show the profiles of state variables such as solvent saturation, vapor pressure, and temperature within the coating thickness with time.

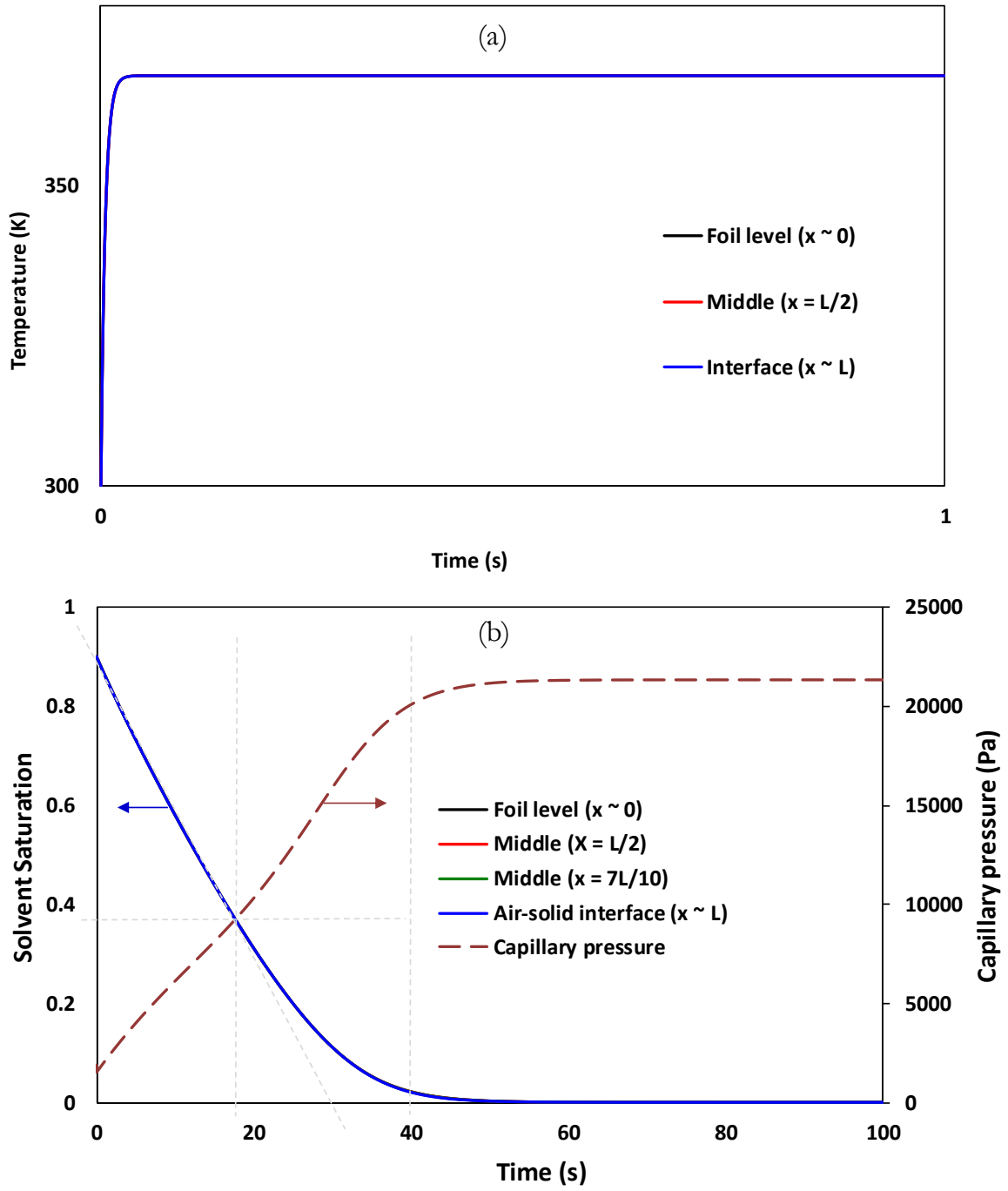


Figure 2 (a) Temperature profile at three different depths of electrode coating (b) Liquid-phase solvent saturation at three different depths of electrode coating

Fig. 2a shows the temperature profile at 3 different points within the coating layer, i.e. air-solid exposed surface ($x = 0$), middle ($x = L/2$), and foil level ($x = L$) against time. We can see that the temperature of the coating across its thickness rises and attains its near final

temperature in a fraction of a second. The thermal conductivity of metal oxides and solvent is high enough that the temperature across the thin coating of electrode is same at any time. This also suggests that the heat transfer is very fast within the coating and so, it cannot be a rate limiting factor for the drying process.

Now the evaporation and the transport of liquid-phase solvent (because of capillary, gravity, and convective forces) are the causes for decrease in liquid phase solvent content within the coating. Recall that we assumed for our model the liquid and vapor phases of solvent are continuous and homogeneously distributed initially in the coating. This follows that the evaporation can happen simultaneously at all locations within the coating. The physical interpretation for this is that the evaporation will initially be rapid in the larger pores and then in the smaller ones. As discussed earlier, the heat transfer is fast and the coating attains its near final temperature across its thickness in the first few seconds so, one may assume that the heat transfer is not a rate limiting step for drying. Consequently, we may expect that the drying rate, if restricted, may be governed by the mass transfer coefficient at the exposed interface and by the equilibrium vapor pressure of solvent in the interior of the coating. Fig. 2b shows the change in liquid-phase solvent saturation (S) and capillary pressure as a function of time at different heights of the coating (foil level, middle, and exposed interface). We can see in Fig. 2b that the slope of solvent saturation profile in the coating decreases almost linearly with time until about 20 s (or $S \approx 0.4$). If the rate of solvent removal continued at this rate the drying will be completed in about 30 s. However, after around 20 s, the rate of change of solvent saturation in the coating begins to slow down and more significantly so after about 30s. One of the main reasons behind this is that the capillary pressure increases sharply until about 20 s as observed in Fig. 2b. This assists in

liquid-phase solvent flux and consequently in its evaporation. After 30 s, the increase in capillary pressure is impeded as it approaches its maximum value where, the liquid pressure becomes constant. As the rate of increase of capillary pressure decreases and becomes negligible, the liquid phase transport is impeded and results in its non-uniform distribution across the coating. This reduces the rate of evaporation and consequently reduces the rate of change of liquid saturation. Also, we can observe from Fig. 2b that the liquid-phase solvent across the coating thickness decreases almost simultaneously. Drying is completed in about 54 s, i.e. about 99.9% of solvent is removed in 54 s of operation.

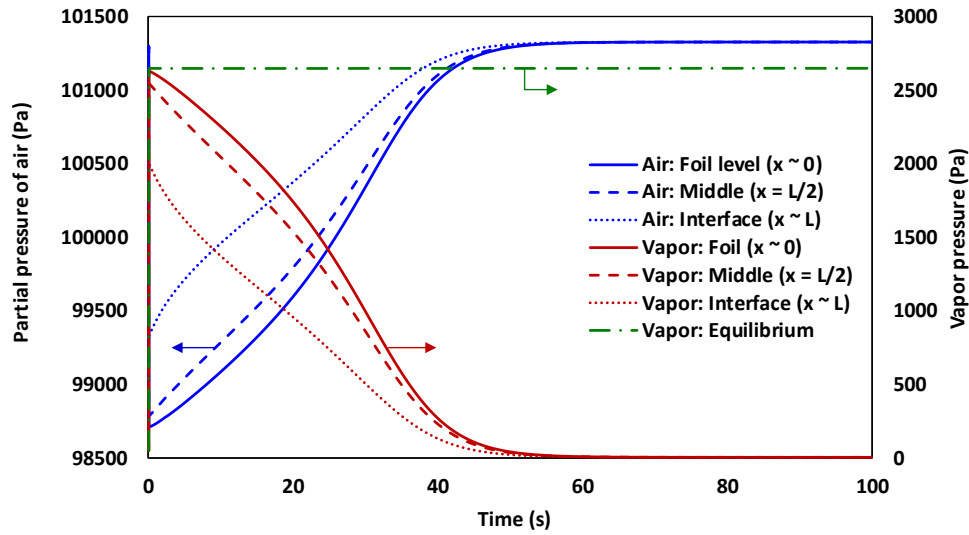


Figure 3 Partial pressure of air and solvent vapors at three different depths and equilibrium vapor pressure in electrode coating

Initially, the void space in the coating is almost filled with liquid-phase solvent and a small fraction of this void space consists of air and solvent vapors at equilibrium. The total pressure in the coating is then equal to the sum of partial pressures of vapor and air and is almost always equal to the atmospheric pressure. Fig. 3 shows the change in partial pressure of solvent and air at different heights of the electrode coating (i.e. foil level, middle, and

exposed interface) with time. Initially, as discussed earlier, the partial pressure of solvent vapors is in equilibrium with the ambient air. As the evaporation of solvent starts, the partial pressure of solvent increases with temperature. The equilibrium vapor pressure increases rapidly along with temperature and reaches its maximum value (within a second). Also, the partial pressure of solvent vapors increases with temperature along with its equilibrium value. The vapor pressure is seen to fall off as a result of convection and diffusion of the vapor out of, and air influx into, the pore volume. As shown in Fig. 3, the total pressure is maintained. The total pressure may slightly change momentarily during evaporation but then it automatically adjusts to the ambient pressure by adjusting the partial pressures of air and vapor. The plots for the partial pressure at a given height (h) shows the net effect of vapor transport (in and out of the differential thickness). Initially, the evaporation and vapor transport is slower near the foil of the coating as compared to the coating layer immediately above it and so on. As observed from Fig. 3, the vaporization is relatively faster at the solid-air interphase as compared to the foil-level. This is because of the lower resistance of mass transport due to shorter transport path (distance to interphase) near the interphase as compared to the foil level. Again, the increased partial pressure with the depth of coating is mainly due to the similar reason. The drying air just above the exposed layer of coating sweeps the evaporating solvent vapors almost immediately. We ensure this through our boundary condition on the vapor pressure at the exposed surface.

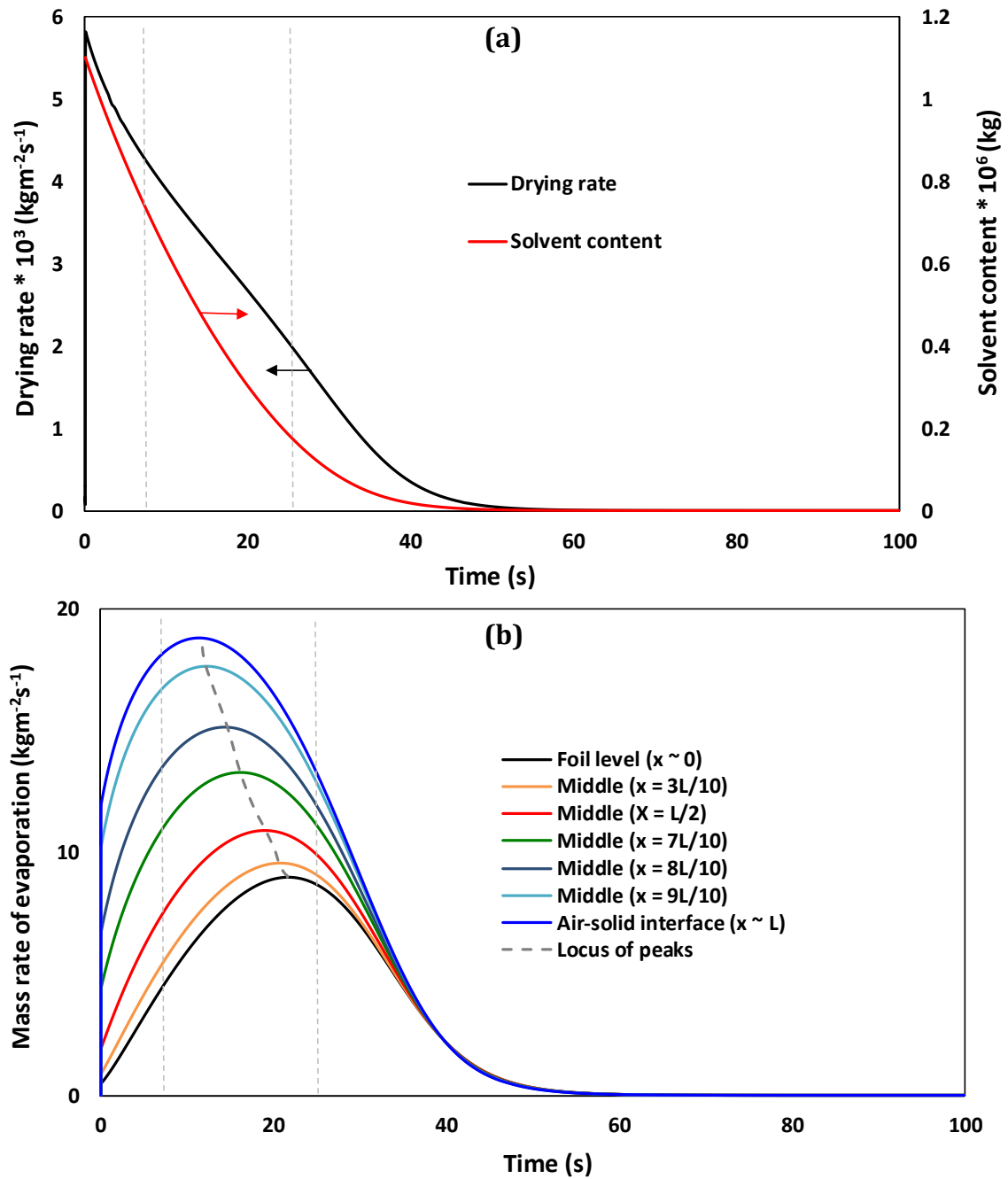


Figure 4 (a) Solvent drying rate and total solvent inventory in the differential volume; (b) Change in rate of evaporation with time at different coating thickness

Fig. 4a shows time profile of the overall drying rate (or solvent removal rate) on mass basis and the total amount of solvent (vapor and liquid) remaining in the coating. Fig. 4b shows the change in mass rate of evaporation with time in the differential volume at different

positions of coating thickness. Fig. 4a shows three distinct zones of varying drying rate. In the zone 1 (0 – 10 s), the drying rate changes rapidly and consequently the reduction of solvent content is fast. This period of 0 – 10 s can be observed in Fig. 4b, where the rate of evaporation across the coating depths is increasing rapidly. The zone 2, i.e. 10 – 25 s, Fig. 4b shows that the rate of evaporation slows down across the thickness and goes through a maximum. This is reflected in the change of solvent amount and drying rate as seen in Fig. 4a. From 10 s to nearly 25 s (i.e. zone 2), the solvent content and drying rate reduces almost linearly with time and at a relatively lower rate as compared to the zone 1. After 25 s (i.e. zone 3), the evaporation rate decreases across the coating thickness at an almost similar rate as seen in Fig. 4b. The drying rate decreases in zone 3 sharply and becomes a very slow process. Most of the liquid-phase solvent has been vaporized after 30 seconds. The rate controlling step then is the removal of vapor phase solvent. Here (after about 30 s), the drying rate becomes very slow and is essentially governed by the vapor phase transport, i.e. air-vapor diffusion process. We consider the electrode to be dried if at least 99.9% of the solvent is removed. From Fig. 4a, we see that almost 50% of the solvent is removed in ~10 seconds, and 90% is removed in 30 seconds. The drying is completed in about 54 s. This shows that nearly half of the total drying time (24 out of 54 s) is needed to remove the last 10% of solvent from the coating. Note that in Fig. 4a the drying rate increases very quickly as the drying begins and peaks at the same time as the temperature reaches its maximum value. This initial peak of drying rate represents the solvent flux through the coating surface, which at a higher rate may potentially lead to stress related defects or cracks in the electrode. Thus, it is important to study the effect of drying rate on the coating quality. More importantly, a longer tail or the slower region of drying rate (i.e. more than half of the total

drying time to dry around only 10% of the remaining solvent) is also the most inefficient region of drying. In other words, a lot of energy is wasted during the slower rate period of drying. In addition, low drying rates translate to longer processing time of electrodes and thus, requiring large dryers. Both of these qualities are undesirable for an electrode manufacturing industry. Accelerating the last stage (or the slower region) of drying can reduce the size of the dryer, eliminate a potential process bottleneck in electrode manufacturing, and lower the overall energy cost.

Effect of process parameters on drying

In order to expedite the drying process and make it more efficient (especially in the latter half, where the drying rate is very slow), we first need to understand its sensitivity to the process parameters. For this, we study the sensitivity of drying rate to the drying temperature and air velocity over the coating.

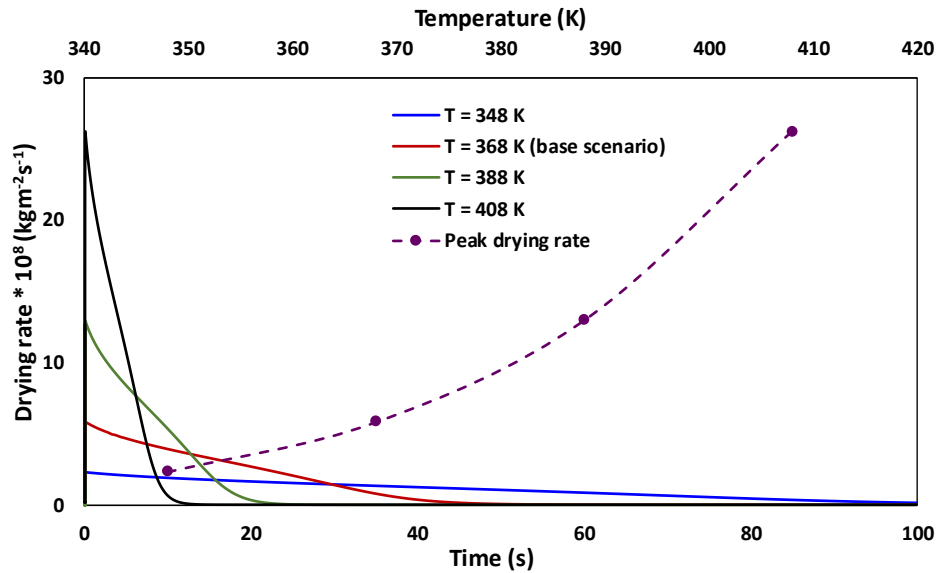


Figure 5 Solvent drying rate profiles at different temperatures of drying medium (hot air)

Fig. 5 shows overall drying rate of the electrode coating at 4 different temperatures ($T = 408, 388, 368$, and 348 K) of the drying medium (i.e. hot dry air). The drying times (or the

time to remove 99.9% of solvent from the coating) are 133 s, 54 s, 24 s, and 12 s for the hot temperatures of $T = 348$ K, 368 K, 388 K, and 408 K, respectively. Clearly, as we increase the temperature of the heating medium, the drying of electrode is expedited significantly. The drying rate profiles for all different temperatures follow a similar trend as discussed earlier in the base case scenario ($T = 368$ K). However, the peak rate of drying increases rapidly with temperature. As shown in Fig. 5, the peak drying rate for $T = 348$ K is about $2.3\text{E-}8 \text{ kgm}^{-3}\text{s}^{-1}$ and for $T = 408$ K is around $2.6\text{E-}7 \text{ kgm}^{-3}\text{s}^{-1}$. So, the chance of developing stress related defects or cracks caused by a high flux of solvent increases with increase in temperature. At lower temperatures (e.g. $T = 348$ K), the drying rate profile is relatively flatter and takes much longer to dry the electrode (133 s). Thus, to reduce the drying rate peaks require drying at lower temperatures but at the cost of increasing drying time. This suggests that a combination of low and high drying rates may be useful in maintaining a lower peak for drying rate (i.e. maximum solvent flux) and at the same time reduces the overall drying time.

Apart from temperature, mass transfer coefficient affects the electrode drying rate. So far, we have specified mass transfer and heat transfer coefficients as fixed parameters in our model ($k_m = 1\text{e}^{-1} \text{ ms}^{-1}$, $k_T = 100 \text{ WmK}^{-1}$). In order to understand the effect of heat and mass transfer coefficients on drying rate, we now consider them as functions of temperature, and velocity of hot dry air flowing over the electrode coating. To compute the heat transfer coefficient, we use the following expression for the turbulent flow of hot air over a flat slab.

$$k_T = 0.037 V_a^{0.8} \left(\frac{\mu_a}{\rho_a} \right)^{-0.8} Pr^{1/3} \lambda_a L^{-0.2}$$

where, V_a is the flow velocity of the hot dry air over drying electrode, μ_a is viscosity of air, Pr is the Prandtl number of drying air, λ_a is the thermal conductivity of air, and L is the

characteristic length. Then, we use the following Chilton-Colburn correlation to compute the mass transfer coefficient from the heat transfer coefficient.

$$k_m = \frac{k_T}{\rho_a C_{p_a}} Le^{-2/3}$$

where, Le is the Lewis number.

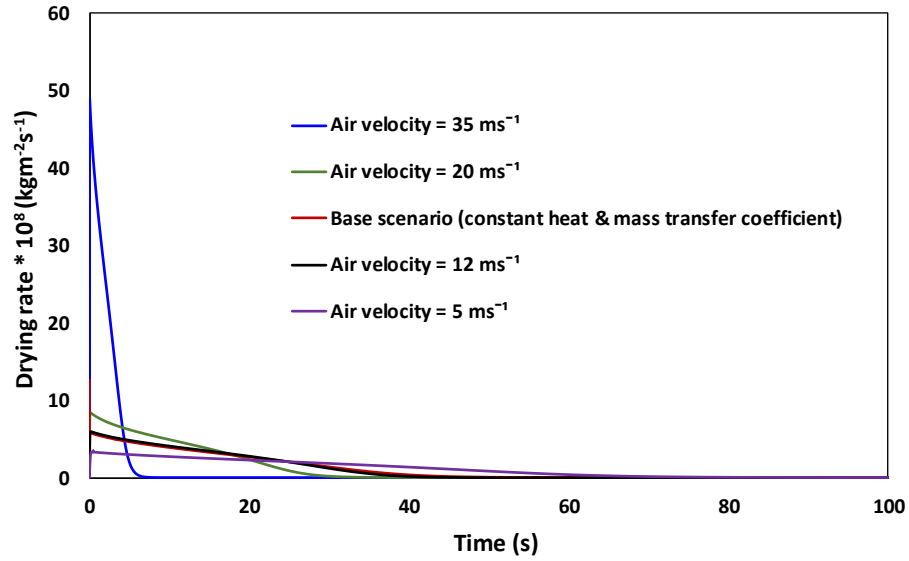


Figure 6 Solvent drying rate at different velocities of hot air flowing above the coating

We now compare the drying rate of electrode coating for 4 different velocities of hot dry air, $V_a = 5, 12, 20,$ and 35 ms^{-1} (Fig. 6). As expected, at higher velocities of air, electrode drying (99.9% of solvent removal) is expedited, e.g. for $V_a = 5, 12, 20,$ and 35 ms^{-1} the drying times are 88, 47, 33, and 6.5 s, respectively. Note that the drying rate profiles in Fig. 6 are similar to that in Fig. 5. In other words, the effect of increasing air velocity is same as increasing temperature on the drying rate. Similar to the base case scenario, the drying rate first reduces rapidly and then slows after nearly 90% solvent is removed. As we reduce the speed of hot air over the coating, the drying rate profiles becomes relatively flatter and spreads relatively more evenly across the drying time. In other words, the peak solvent flux value

during drying, which is attained very quickly, increases significantly with increase in the velocity ($3.5 \text{ kgm}^{-3}\text{s}^{-1}$ for $V_a = 5 \text{ ms}^{-1}$ and $49 \text{ kgm}^{-3}\text{s}^{-1}$ for $V_a = 35 \text{ ms}^{-1}$) of the air flowing over the electrode. These observations can now be used to design a more efficient and effective drying rate profile by adjusting the process parameters such as temperature and hot air velocity.

Multi-stage drying operation

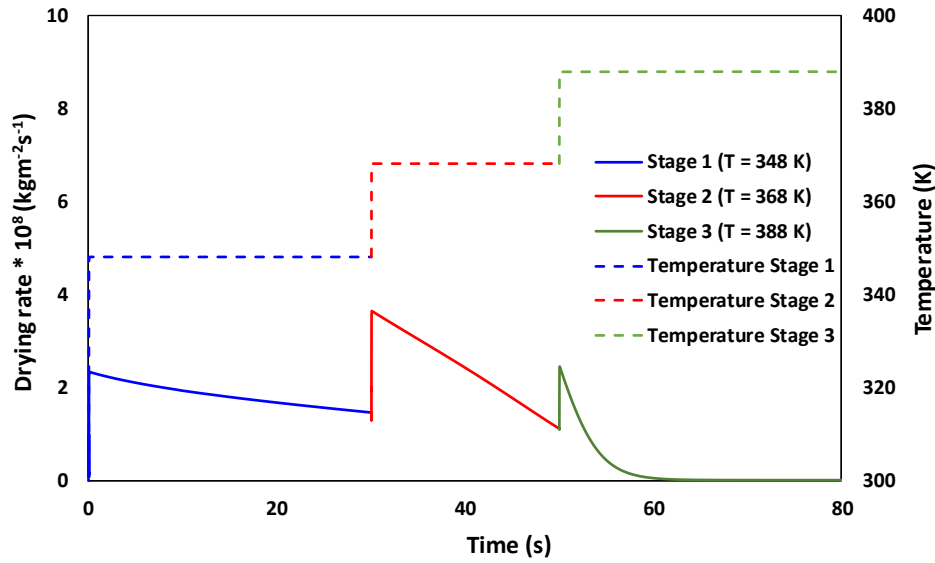


Figure 7 Drying rate and temperature profile in a 3-zone dryer with stage 1 at 348 K, stage 2 at 368 K, and stage 3 at 388 K

As mentioned earlier, expediting the drying process can potentially reduce the size of dryer and decrease the energy demand but may impact the quality of electrode due to the formation of cracks and the additional stress induced from the high drying rates. Thus, the drying rate needs to be appropriately designed to both reduce the processing time and not affect the product quality. One of the popular methods is to divide the drying period into multiple zones of varying temperatures or multi-zone drying. Fig. 7 shows the temperature and drying rate profile in such a multi-zone drier with 3 zones of increasing temperatures, i.e. zone 1 at $T = 348 \text{ K}$, zone 2 at $T = 368 \text{ K}$, and zone 3 at $T = 388 \text{ K}$. The scope of this example

is only to understand the effect of multi-zone drying and so, the durations and temperatures in the 3 zones are not optimized. However, the observations from this example can be used to strategize and optimize the drying process by ensuring appropriate duration and temperature for each zone. From Fig. 7, we know the drying time with $T = 348\text{K}$ is ca. 130 s, with $T = 368\text{ K}$ is about 54 s, and with $T = 388\text{ K}$ is about 24 s. In the 3-zone drier, the total drying time is around 60 s with 30 s at $T = 348\text{ K}$, 20 s at $T = 368\text{ K}$ and 10s at $T = 388\text{ K}$. Assuming the flow rate of air to be constant, the aforementioned scheme of multi-stage drying reduces the energy consumption by ca. 50% as compared to the base case scenario discussed earlier in this paper. In addition, the peak solvent flux or the peak drying rate in the 3-zone drier scheme is achieved in the zone 2 with $T = 368\text{ K}$ and is ca. $3.6\text{ kgm}^{-3}\text{s}^{-1}$. This is again about 40% less than the base case scenario, where the peak drying rate is around $5.8\text{ kgm}^{-3}\text{s}^{-1}$. This example shows an approach to reduce the overall energy required for drying by distributing drying period over a range of temperatures and to decrease solvent flux induced defects on the electrode coating due to high solvent drying rates. Also, one can optimize the zonal drying duration and temperature to reduce the overall heat load by reducing the amount of air flow in the high temperature zone and increasing a proportional air flow in the low temperature zone.

Effect of solvent selection on drying

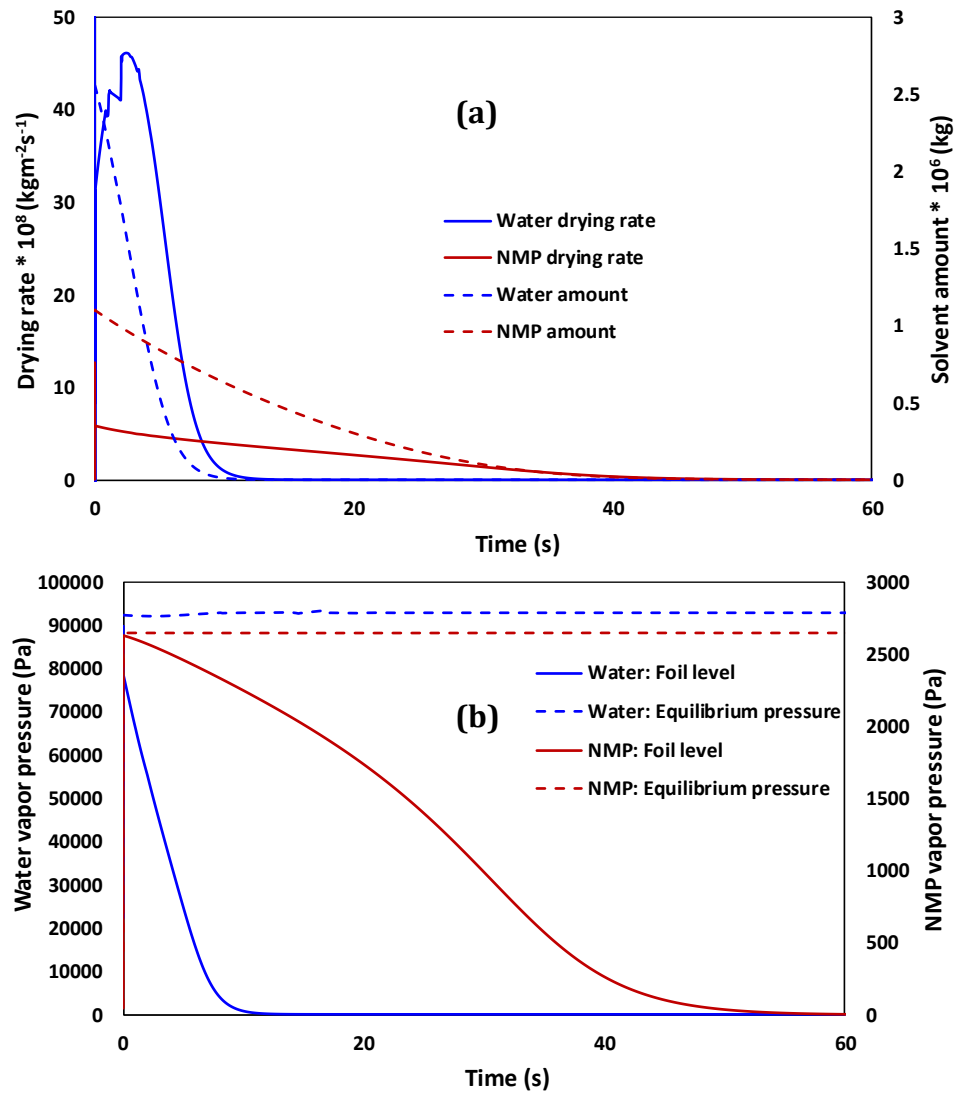


Figure 8. (a) Drying rate profile and total solvent amount for NMP and Water; (b) Vapor pressure during drying near foil level for NMP and Water

Table 2. Comparison of NMP and water properties

Solvent	T = 368.15 K	NMP	Water
Surface tension	Nm^{-1}	6.13E-05	6.14E-05
Enthalpy of vaporization	kJkg^{-1}	513	2257
Specific heat capacity	$\text{kJkg}^{-1}\text{K}^{-1}$	1.9	4.2
Equilibrium vapor pressure	Pa	2654	83768

Finally, we study the effect of using different solvents such as an aqueous solvent (water) instead of NMP, on the process of electrode drying. The physical properties of water and NMP and the process parameters of drying are listed in Table 2. Note that the heat of vaporization for water is nearly 4.5 times that of NMP on a mass basis. Also, specific heat capacity of water is much higher than that of NMP. Thus, intuitively, one would expect that the total energy and time required to remove water from the coating would be much higher than that of NMP. However, the equilibrium vapor pressure for water is nearly 35 times that of NMP as shown in Fig. 8a, ca. 93000 Pa for water versus ca. 2653 Pa for NMP at $T = 368.15$ K. This means that a unit volume of drying air can ideally remove 35 times more of water vapor as compared to NMP vapors. As the higher equilibrium vapor pressure will provide a higher driving force for faster evaporation of water so, the water based electrode coating will dry much faster than NMP-based coatings. Fig. 8b clearly shows that water dries nearly 4.5 times faster than NMP (54 s for NMP versus 12 s for water). Although the energy required for evaporation will be higher for water, as it evaporates faster than NMP, the overall energy required for the drying will be lower for water. Assuming an identical flow rate of hot dry air, the energy consumption for drying electrodes can be compared based on (1) the time of drying time, and (2) the amount of solvent removed. Fig. 8b gives the information regarding drying time and the amount of solvent can be calculated from the procedure shown in the formulation section and the data in Table 2 (i.e., solids%, density, and saturation). In this study, we observe that the total energy required to dry water is nearly 10 times less that required for NMP.

Remarks

Any model is only as good as the values of parameters used in the model. The aforementioned model and analysis, present an understanding of the drying process within the electrode coating and identify factors affecting it. The results we presented are good within the assumptions we made during the model development.

As mentioned earlier, we focused on the drying in the second phase regime with $S < 1$, whereas overall drying process may also involve the constant drying rate phase. Although such a phase would relatively be much shorter than the second phase because of the various forces acting on the solvent during drying in the second phase. Thus, the drying time we presented in this study may under estimate the overall drying as compared to the actual drying process. Although, in our study, we did not consider heat loss at the impermeable boundary of the coating due to the heat capacity of current-collector foil, we have observed that such losses are negligible compared and do not impact our results. Furthermore, in a two-sided coater and drier such effects will be further minimized.

In addition, in our model, we assumed the values for parameters such as heat transfer coefficient and mass transfer coefficient. These parameters have a significant impact on the results. The actual values for these parameters could be measured from experiments. In absence of such data, our results are limited to the accuracy of the chosen values.

Conclusions

We presented an analysis of the electrode drying process in the context of lithium ion batteries using a continuum level mathematical model. The mass transport and heat transfer in the electrode coating during drying were studied in detail. The results showed rapid temperature rise suggesting that the electrode drying is not a heat transfer controlled but a mass transfer controlled process. One of the main reasons for this is the very thin electrode

layer (thickness $\approx$ 150 microns). Also, the heat transfer coefficient for forced convection using hot dry air over the thin coating is high and the heat required to evaporate the small quantity of solvent is not high. The high thermal conductivity of metal oxides and solvent is high enough that the temperature across the coating thickness is the same.

For the 150 micron thick electrode coating the solvent disappears (or dries) rapidly in the first 30s and then slows down because of mass transfer limitations. The solvent drying profiles from our analysis suggests that about 90% of drying is completed in nearly half of the total drying time. This means that nearly half of the drying time is dedicated only to remove the last 10% of the solvent. This leads to the wastage of a large portion of drying energy and presents a great scope for process optimization to improve process efficiency. In addition, the initial drying rate or solvent flux out of the electrode coating is very high and then reduces with time. This high drying rate or solvent flux can potentially lead to stress related defects or form cracks in the electrode coating. In order to strategize mitigation of aforementioned issues, we have presented an analysis of the effect of various process parameters (temperature and air velocity) on solvent drying rate. Higher temperatures favor faster drying. Increasing the air velocity increases the mass transfer coefficient thereby accelerating the drying rate.

Based on aforementioned observations, it is possible to design an optimal drying profile by using a multi-zone dryer, where temperatures, residence time, and air velocity in each zone can be independently controlled. Our results indicate that a 3-zone drying of electrode reduces the drying time as compared to drying in single zone with a lower temperature. We further estimate that at least 50% of the overall energy demand can be reduced by carefully choosing zonal durations and temperatures. Also, we show that the peak solvent flux can be

reduced by at least 40% by drying electrodes in a multi-temperature dryer along with reducing overall drying time and energy.

Furthermore, we study and compare the drying behavior of electrodes using an aqueous solvent (i.e. water) instead of NMP. Despite the higher heat of vaporization (per mass) and specific heat capacity of water, it can be dried 4.5 times faster than NMP. The energy required to vaporize the water solvent is relatively higher but the shorter drying time results in lower overall energy required for the process using water as solvent. For the conditions reported here, the drying of water requires nearly 10 times less energy than for NMP.

Finally, the study presented in this paper highlights the process dynamics of drying lithium-ion battery electrodes, its sensitivity to the changes in the process parameters, and suggests strategies to expedite drying and minimize potential defects. Thus, this work lays a good foundation for future studies on process optimization and designing optimal drying profiles for battery electrodes.

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